Pure
Flash
Array

Pure Storage
FlashArray Administrator

PSFA-Admin Interview Prep Guide

300 Questions, Answers & Real-World Examples

FlashArray | Purity OS | ActiveDR | SafeMode | Pure1 | VMware

- FlashArray Architecture
- Purity//FA Administration
- Volume & Host Management
- Replication & DR
- VMware & vSphere Integration
- ActiveDR & ActiveCluster
- Pure1 & Analytics
- Performance & Troubleshooting

Akhilesh Chaudhari | TechShiksha
@akhilesh.webdev | akhileshchaudhari.com

TABLE OF CONTENTS

SECTION 01 | FLASHARRAY ARCHITECTURE & PURITY OS FUNDAMENTALS

Q1 What is Pure Storage FlashArray and what makes it different?

A Pure Storage FlashArray is an all-NVMe, all-flash storage array built for enterprise workloads. Key differentiators: (1) Purity//FA operating system with always-on inline deduplication, compression, and thin provisioning. (2) DirectFlash modules — proprietary NVMe flash eliminating SSD controllers for lower latency. (3) Evergreen storage model — non-disruptive controller upgrades without data migration. (4) Pure1 cloud management portal. (5) No performance tiers — all flash, always full performance. Active-active dual controller HA.

Example: EMC VNX requires tiering configuration (SSD, SAS, NL-SAS). NetApp FAS requires volume efficiency policies. Pure FlashArray: provision a volume — dedup, compression, thin-provisioning active by default, no configuration needed. New DBA provisions Oracle LUN in 30 seconds via GUI, immediately gets full array performance. Simplicity is the core differentiator.

Q2 What is the Pure Storage FlashArray hardware architecture?

A FlashArray hardware: (1) Dual controllers (active-active HA) — both serve I/O simultaneously. (2) DirectFlash Modules (DFMs) — proprietary NVMe flash in back of chassis. (3) Shared memory between controllers via NVMe-oF fabric. (4) DirectMemory Cache (DMC) — DRAM write cache with NVRAM protection. (5) Front-end ports: Fibre Channel (16/32Gbps), iSCSI (10/25GbE), NVMe-oF. (6) Unified backend for dual-controller shared storage access. Models: //X (performance), //C (capacity), //E (entry).

Example: FlashArray//X90 R3: 2 controllers, 24 DirectFlash Modules (up to 2.4PB raw). Both controllers active — writes acknowledged when written to NVRAM on both controllers. Controller failure: partner serves all I/O with zero data loss in <30 seconds. Front-end: 4x32Gbps FC per controller + 4x25GbE iSCSI per controller.

Q3 What is Purity//FA operating system?

A Purity//FA is Pure Storage OS for FlashArray. Features: inline deduplication and compression (always on, no configuration), thin provisioning, Snapshots, replication (async, sync, ActiveCluster, ActiveDR), REST API 2.x, Pure1 integration, automatic non-disruptive software upgrades. Major versions: Purity 5.x (mainstream), 6.x (NVMe-oF, ActiveDR, SafeMode enhancements). Upgrades are typically non-disruptive — one controller at a time while partner serves I/O.

Example: Purity 6.1 upgrade from 5.3: Pure1 recommends upgrade. Non-disruptive: CT1 reboots with new Purity (CT0 serves I/O for 8 minutes), then CT0 reboots. Total: 20 minutes, zero host I/O interruption. New features enabled: NVMe/RoCE connectivity, enhanced ActiveDR. Purity upgrades are the simplest enterprise storage OS upgrades available.

Q4 What is DirectFlash technology?

A DirectFlash is Pure's proprietary NVMe flash module eliminating the SSD controller layer. Traditional SSD: NAND + SSD controller introduces write amplification and garbage collection pauses. DirectFlash: Purity OS manages NAND directly — global wear leveling across all modules, no per-SSD garbage collection storms, lower write amplification, consistent ultra-low latency. DirectFlash enables Purity to optimize flash management at the global array level instead of per-module.

Example: Traditional SSD-based array: garbage collection causes latency spikes to 2-5ms every few minutes. DirectFlash: Purity manages NAND globally — no per-module spikes. Oracle DBA sees consistent 0.2-0.5ms latency vs competitor 0.2ms average with 5ms spikes. For OLTP: P99 latency consistency matters more than average — DirectFlash wins on consistency.

Q5 What is Purity//FA data reduction?

A Purity//FA efficiency: (1) Thin provisioning — only written blocks consume physical space. (2) Inline deduplication — identical 512-byte fingerprint blocks stored once globally. (3) Inline compression — remaining unique blocks compressed. (4) Pattern elimination — all-zero blocks not stored. All inline, always on, zero admin configuration. Typical ratios: VMware 5-10:1, Oracle 2-3:1, VDI 10-20:1. Pure guarantees minimum 4:1 data reduction in contract — if actual ratio below 4:1, Pure provides free capacity.

Example: VMware: 100 Windows VMs, 50TB provisioned. Pure dashboard: dedup 3.2:1, compression 1.8:1, total 5.7:1. Physical used: 8.8TB. Guarantee: if ratio falls below 4:1, Pure adds capacity at no charge. Customer buys 25TB raw to provide 100TB effective at 4:1. Data reduction guarantee eliminates capacity purchase risk.

Q6 What is Pure Storage SafeMode?

A SafeMode (immutable Snapshot protection): enabled Snapshots cannot be deleted or modified until retention period expires — even by array administrators. Three-level authentication required to modify SafeMode settings (array admin + Pure Support + Pure CSM). Protects against: ransomware deleting backups, insider threat, compromised credentials. Configure: retention period per protection group (minimum 24 hours). SafeMode Snapshots cannot be destroyed with standard commands — 24-hour eradication delay after retention expires.

Example: Ransomware encrypts volumes and attempts to delete SafeMode Snapshots. 'purepgroup destroy --eradicate prod_pg.snap' fails: 'SafeMode protection active.' Attacker cannot delete protected Snapshots. Admin calls Pure Support, 3-factor authentication, verifies incident. SafeMode Snapshot from before attack restored. Volume overwritten with clean Snapshot — data recovered. SafeMode is the most critical ransomware defense feature on FlashArray.

Q7 What is Pure Storage Evergreen Storage model?

A Evergreen storage: (1) Evergreen//Forever — annual subscription, new controller hardware every 3 years (non-disruptive swap). (2) Evergreen//One — storage-as-a-service, Pure owns hardware, customer pays per TB consumed. Non-disruptive controller upgrade: new controller slides in, data non-disruptively migrated, old controller removed. No data migration project, no downtime, always current generation. Addresses hardware obsolescence without disruptive refresh cycles.

Example: Traditional: buy EMC VMAX, year 5 migration project (6 months, $500K consulting, 2 outages). Pure Evergreen: year 3, Pure delivers new controller generation. Non-disruptive swap: 4 hours, array now latest generation. Year 6: another refresh. No migration projects ever. Total cost of ownership: 40% less than traditional 5-year refresh cycle.

Q8 What is FlashArray//X vs //C vs //E product line?

A FlashArray//X: performance-optimized, DirectFlash NVMe (TLC), <0.5ms latency, highest IOPS, for OLTP/VDI/AI. FlashArray//C: capacity-optimized, QLC flash, 50% lower cost/TB than //X, for secondary storage and warm data. FlashArray//E: entry-level, lower cost, for SMB and remote offices. All run Purity//FA, same management, same replication. Mix: //X for production, //C for dev/test — unified management via Pure1. Same APIs and automation work across all models.

Example: Enterprise strategy: Oracle OLTP → //X90 (NVMe, <0.5ms). VMware general → //X50 (performance). Dev/Test → //C60 (QLC, 50% lower cost/TB). All managed via Pure1 single pane of glass. //X to //C replication for dev refresh. Unified platform, differentiated cost tiers matching workload requirements.

Q9 What is Pure Storage ActiveCluster?

A ActiveCluster: synchronous replication between two FlashArrays, zero RPO, transparent failover. Both arrays serve I/O simultaneously — active-active stretch cluster. Every write acknowledged only when written to both sites. On site failure: surviving array serves 100% I/O, hosts MPIO auto-reroutes, no admin intervention. Uses Mediator (third site tiebreaker) for split-brain resolution. Requirements: <10ms RTT, Mediator, Purity 5.0+. Hosts connected to both arrays via multipathing.

Example: NYC and DC2 5km apart, ActiveCluster. Oracle RAC: half nodes NYC, half DC2, all connected to both arrays. DC1 loses power: DC2 FlashArray serves 100% I/O, Oracle nodes in DC2 continue. RPO=0 (synchronous), RTO<30 seconds (Oracle node restart). Zero data loss. Mediator on AWS confirms DC1 unreachable before DC2 promotes.

Q10 What is Pure Storage ActiveDR?

A ActiveDR (Purity 6.1+): async replication with always-ready DR volumes and fast RTO (<5 min). DR volumes always mounted read-only on DR array as virtual shadow volumes. On failover: no 'break' needed — one Promote command makes volumes instantly writable. Hosts at DR site pre-configured, start immediately. RPO configurable to 60 seconds minimum. Compare: traditional async failover 15-30 minutes vs ActiveDR <5 minutes.

Example: SQL Server protected by ActiveDR, NYC→Chicago, 5-minute RPO. Chicago FlashArray: SQL volumes as demoted shadow copies (auto-refreshed every 5 minutes). Failover: 'purepgroup promote prod_pg' — 30 seconds. Chicago SQL hosts: volumes writable. SQL starts: 3 minutes. Total RTO: 4 minutes vs traditional 15+ minutes. Pre-staged hosts = major RTO improvement.

Q11 What is Pure1 cloud management?

A Pure1: SaaS management and analytics platform for all FlashArrays. Capabilities: (1) Array management from browser (provision volumes, hosts, connections). (2) Global fleet — all arrays in one dashboard. (3) Pure1 Meta — AI/ML workload analysis, anomaly detection. (4) Capacity planning — ML-based forecasting. (5) Upgrade Advisor — automated upgrade scheduling. (6) Proactive support — Pure1 detects issues, auto-opens cases. (7) Sustainability tracking. Data from array phone-home telemetry.

Example: Pure1 proactive support: ML detects wear on DFM3 — predicted failure in 14 days. Auto-opens support case. Pure engineer calls customer: 'We detected an issue, field engineer scheduled Thursday.' DFM replaced before failure. Customer sees zero storage incident. This proactive model is Pure's strongest customer satisfaction differentiator — support that resolves issues before customers notice.

Q12 What is FlashArray REST API?

A FlashArray REST API 2.x: JSON-based, Bearer token authentication. SDKs: py-pure-client (Python), purestorage PowerShell. Key endpoints: /api/2.x/volumes, /api/2.x/hosts, /api/2.x/connections, /api/2.x/protection-groups. Ansible: purestorage.flasharray collection. Terraform: purestorage provider. All GUI operations available via API — complete automation. Rate limit: 100 req/sec per token. TLS 1.2+ required.

Example: Python: 'import purestorage; array = purestorage.FlashArray(IP, api_token=TOKEN); array.create_volume(vol_name, size)'. Ansible: 'purefa_vol: name: oracle_data size: 2T state: present'. ServiceNow → webhook → Python → volume provisioned in 10 seconds → ticket resolved. Zero manual storage admin steps for standard provisioning.

Q13 What is FlashArray front-end port configuration?

A FlashArray ports: FC ports (CT0.FC0-FC3, CT1.FC0-FC3) — 16 or 32Gbps. iSCSI/NVMe ports (CT0.ETH0-ETH5, CT1.ETH0-ETH5) — 10/25GbE. Each port belongs to a controller. Both controllers active — hosts connect to ports on both CTs for HA. FC: configured as target N_Ports, WWPN used for zoning. iSCSI: IP address configured per port. Connect hosts to CT0 AND CT1 ports — survives controller failure.

Example: //X50: CT0.FC0, CT0.FC1 + CT1.FC0, CT1.FC1. Host HBA0 → FabA → CT0.FC0. Host HBA1 → FabB → CT1.FC0. 2 paths per volume. CT0 fails: CT1.FC0 path remains — all I/O continues. ALUA: CT0 paths = Optimized, CT1 paths = Non-Optimized normally. After CT0 failure: CT1 paths promote to Optimized.

Q14 What is FlashArray volume provisioning?

A Volume creation: GUI □ Storage □ Volumes □ + New Volume. Enter name and size. Volumes thin by default — no physical reservation. Instant creation regardless of size. Connect to host: Storage □ Hosts □ Connect □ select volume. LUN ID auto-assigned or specified. No RAID config, no aggregate, no pool selection — pure simplicity compared to traditional storage. Snapshot: volume-level instant copy.

Example: Oracle DBA provisions: New Volume 'oracle_data' 2TB (10 seconds). New Volume 'oracle_redo' 500GB (5 seconds). Create Host 'oracle01' with server WWPNs. Connect both volumes. Host FC rescan — volumes appear. ASM adds disks. Total: 4 minutes vs 45 minutes on traditional storage. No storage expertise required for provisioning.

Q15 What is FlashArray host and host group management?

A Host objects represent servers. Host contains: WWPNs (FC), IQNs (iSCSI), NQNs (NVMe). Host Group: logical grouping of hosts sharing volume access — equivalent to igroup with multiple initiators. Volumes connected to host group visible to all hosts in group. Create: Storage □ Hosts □ New Host □ add initiators. Connect volume to host or host group. Best practice: host groups for clusters (ESXi, Oracle RAC). Individual hosts for dedicated server access.

Example: VMware cluster 8 ESXi hosts: Host Group 'vmware_cluster'. All 8 ESXi hosts in group. Connect VMFS volume to host group — all 8 see it instantly. New ESXi added: add to host group → inherits all volume connections. No per-host LUN mapping — single host group operation covers entire cluster.

Q16 What is FlashArray protection group and snapshot scheduling?

A Protection Groups: collection of volumes with common Snapshot/replication policy. Create PG ▯ add volumes ▯ set schedule. Snapshot schedule: every 1 minute to daily, configurable retention. Snapshot: instant, space-efficient (only changed blocks). Replication: add target array to PG for async replication. GUI: Protection ▯ Protection Groups. Automatic snapshot creation and retention management — no manual snapshot management.

Example: Oracle PG 'oracle_prod': volumes data, redo, undo, temp. Schedule: every-15min/4hrs, hourly/48hrs, daily/30days. All 4 volumes snapped simultaneously (crash-consistent). DBA drops table at 2:30 PM: restore from 2:15 PM snapshot. 15-minute maximum data loss. Snapshot consistency across related volumes is the key benefit.

Q17 What is FlashArray HA and controller failover?

A FlashArray HA: dual active-active controllers sharing all DirectFlash Modules. Write acknowledgment: both controllers' NVRAM must confirm — zero data loss on controller failure. Controller failover: <30 seconds. No data migration needed. Hosts MPIO reroutes to surviving controller ports. DirectFlash Module access: both controllers share all DFMs via backend. Front-end: host connected to both CT0 and CT1 ports. HA is hardware-level — no separate failover process to initiate.

Example: CT0 crashes. CT1 takes over within 15 seconds. ESXi MPIO: CT0 paths go offline, CT1 paths remain active. VMs: brief I/O pause (<15 seconds), then operational. Pure1: auto-notified, support case opened. Field engineer arrives per SLA. CT0 replacement: slides in, auto-rejoins active-active. Zero data loss, minimal VM disruption.

Q18 What is FlashArray non-disruptive upgrade (NDU)?

A Purity upgrades fully non-disruptive. Process: CT1 reboots with new Purity (CT0 serves I/O, ~8-10 min). CT1 rejoins. CT0 reboots (CT1 serves, ~8-10 min). CT0 rejoins. Both on new Purity. Total: ~20 minutes. Hosts: 2 brief path flaps (one per controller reboot). Applications: no impact with properly configured MPIO. DirectFlash Module firmware also updated non-disruptively. Hardware: DFMs hot-plug, no downtime. Schedule during low-traffic window for cleanest experience.

Example: Purity 5.3.12 → 6.1.4 at 2 AM: CT1 reboots (8 min, CT0 active). CT1 returns. CT0 reboots (8 min, CT1 active). CT0 returns. 20 minutes total. SQL Server: MPIO handles path flaps, zero SQL errors. DBA confirms zero errors in SQL log post-upgrade. Pure NDU is the simplest enterprise storage upgrade process available.

Q19 What is FlashArray performance specifications?

A FlashArray//X performance: <0.5ms read latency (4KB random). //X90 R3: 10M+ IOPS. //X50: 1.5M IOPS. All NVMe-based DirectFlash. Consistent performance — no SSD garbage collection spikes. Key: published IOPS includes inline dedup+compression (many competitors benchmark with efficiency disabled). Real-world 8KB 70/30 Oracle: 650K IOPS at 0.3ms. //C: slightly higher latency due to QLC, 30-50% lower $/TB. //E: entry-level, moderate performance.

Example: Benchmark Oracle OLTP 8KB 70/30 with efficiency on: Pure //X70: 650K IOPS, 0.35ms. Competitor: 420K IOPS, 0.6ms. Pure: 55% more IOPS, 42% lower latency with efficiency active. Always test with production-equivalent workload settings. Pure SPECsfs2014 benchmarks published — performance with real-world workloads.

Q20 What is FlashArray capacity management?

A Capacity concepts: Raw (total DFMs). Provisioned (sum of volume sizes — can exceed physical via thin). Physical used (actual space after dedup/compression). Data reduction = provisioned ÷ physical. GUI: capacity gauge shows physical vs raw. Warning at 75%, critical at 90%. Expansion: hot-plug DFMs, instantly available. Over-provisioning safe until physical approaches 95% — growth plan based on physical, not provisioned.

Example: //X50 50TB raw, 200TB provisioned, 35TB physical used, 2.8:1 reduction. Remaining: 15TB physical. Pure1 forecast: 90% threshold in 6 months. Order DFMs now (8-week lead time). Install hot-plug: raw increases to 80TB. Forecast extends to 18 months. No downtime during expansion.

Q21 What is FlashArray iSCSI network configuration?

A iSCSI configuration: GUI □ Array □ Network □ configure IP on ETH ports. MTU 9000 recommended. VLAN tagging supported. iSCSI target IQN auto-generated: iqn.2010-06.com.purestorage:flasharray-xxxxx. Host IQN added to Pure host object. Multipath: connect to CT0 AND CT1 iSCSI IPs. CHAP: optional per-host in host object. Dedicated storage VLAN recommended. Both controllers serve iSCSI — active-active.

Example: CT0.ETH0 = 10.200.0.10/24, CT1.ETH0 = 10.200.0.11/24, MTU 9000. Windows: iSCSI Initiator → discover 10.200.0.10 → connect. Second path: discover 10.200.0.11 → connect. MPIO: 2 active paths. Volume connected in Pure → appears as new disk. Total setup: 10 minutes.

SECTION 02 | VOLUME MANAGEMENT & HOST CONNECTIVITY

Q22 What is a FlashArray volume and its properties?

A FlashArray volume: block storage LUN equivalent. Properties: name, size (thin by default), serial number (MPIO ID), provisioned size, volume data (logical written), unique data (post-dedup). Volume types: standard (writable), snapshot (read-only point-in-time), clone (writable copy of snapshot). No pools, aggregates, or RAID groups visible to admin — Purity manages internally. Instant provisioning regardless of size. Volumes can be in Volume Groups for consistent snapshot management.

Example: 'purevolume create --size 10T oracle_data' — 10TB created in <1 second. No RAID, no aggregate. Oracle connects, partitions, formats. Compare: EMC LUN requires pool selection, RAID group, metaLUN creation — 20+ steps. Pure volume creation: 3 clicks or 1 CLI command.

Q23 What is a FlashArray snapshot?

A Snapshots: instant, space-efficient point-in-time copies. Creation: milliseconds regardless of volume size. Space: copy-on-write, only stores changed blocks post-snapshot. Snapshots are read-only — copy to create writable volume. Named: volume-name.snapshot-name. From snapshot: create new volume (copy) or overwrite existing (restore). Auto-snapshots via protection group schedules. Manual: one-click in GUI or CLI.

Example: oracle_data (2TB) has snapshot oracle_data.daily.9AM. DBA drops table at 2 PM. Restore option 1: Copy snapshot → new volume oracle_data_recovery (15 seconds). Extract specific table. Option 2: Overwrite oracle_data with 9AM snapshot (instant). 2-minute recovery vs hours from traditional backup. Snapshots are the primary operational recovery tool.

Q24 What is FlashArray volume cloning?

A Clone: writable, space-efficient copy of volume or snapshot. Process: take snapshot □ copy to new volume. Initial copy: metadata only (seconds). Space-efficient: shared blocks with parent, diverges as data changes. Use cases: dev/test from production, pre-upgrade rollback, database refresh. Commands: 'purevolume copy --overwrite source_snapshot new_vol' or GUI □ volume □ Copy. No actual data movement for initial creation.

Example: 5TB production Oracle: Create snapshot (instant). Copy to oracle_dev (seconds — metadata). Dev team modifies extensively. Next quarter: delete oracle_dev, create new from latest snapshot. Physical cost: only modified blocks. 5TB dev environment stored as a few GB of delta. Instant, thin dev environment refresh from production.

Q25 What is FlashArray FC host connectivity?

A FC host config: (1) Create host with server WWPNs. (2) Create host group (for clusters). (3) Connect volume to host. (4) Zone FC switch: host WWPN □ FlashArray CT0.FC0, CT1.FC0 WWPNs. (5) Host rescan □ volume visible. FlashArray ALUA: CT0 paths Optimized, CT1 paths Non-Optimized (or vice versa per volume preference). MPIO routes I/O to Optimized paths normally.

Example: SQL Server FC: Host 'sqlserver01' with 2 HBA WWPNs. Volume 'sql_data' connected (LUN ID auto-1). Zone: HBA0 ↔ CT0.FC0, HBA1 ↔ CT1.FC0. SQL Server rescan → 1TB disk. MPIO: 2 paths. CT0 fails: MPIO routes all to CT1.FC0 path automatically. Zero SQL errors.

Q26 What is ALUA on FlashArray?

A FlashArray ALUA: both controllers active-active, each volume has preferred controller. Optimized paths: via preferred controller (lowest latency). Non-Optimized paths: via secondary controller (slightly higher latency, works correctly). Hosts with MPIO use Optimized paths primarily. Controller failover: ALUA state changes, surviving controller's paths become Optimized. No host reconfiguration needed. ALUA ensures correct path selection automatically across all protocols (FC, iSCSI, NVMe).

Example: Volume 'oracle_data' preferred on CT0. CT0.FC0 and CT0.FC1 = Active/Optimized (prio=50). CT1.FC0 and CT1.FC1 = Active/Non-Optimized (prio=10). Normal: I/O via CT0 (lower latency). CT0 fails: ALUA transitions CT1 paths to Optimized. Host MPIO switches automatically. After CT0 recovery: rebalances.

Q27 What is FlashArray iSCSI host connectivity?

A iSCSI host config: (1) Configure iSCSI IPs on array ETH ports. (2) Create host object with host IQN. (3) Connect volume to host. (4) Host iSCSI Initiator: add discovery portals (CT0 IP + CT1 IP). (5) Login to target. (6) MPIO configured for 2 paths. CHAP: optional, configured per host in Pure host object. MTU 9000 recommended. Both CT0 and CT1 iSCSI paths needed for HA.

Example: Linux iSCSI: 'iscsiadm -m discovery -t sendtargets -p 10.200.0.10' → discovers Pure IQN. 'iscsiadm -m node -l' → sessions established. Second path: discovery to 10.200.0.11 (CT1). 'multipath -ll' shows 2 paths. Volume connected in Pure → /dev/sdb on Linux. Mount and use.

Q28 What is FlashArray NVMe-oF connectivity?

A NVMe-oF (Purity 6.1+): NVMe/FC and NVMe/RoCE. Host NQN added to Pure host object. FlashArray presents NVMe namespaces. NVMe/FC: same FC fabric, same zoning. NVMe/RoCE: RDMA over 100GbE DCB. 'nvme connect -t fc' or 'nvme connect -t rdma'. Volumes appear as NVMe namespaces: 'nvme list'. Latency advantage: 50% lower than SCSI protocols on same hardware.

Example: Linux NVMe/FC to FlashArray: Zone host HBA WWPN ↔ FlashArray FC WWPN. Pure host object: Linux NQN added. 'nvme connect-all' discovers namespace. 'nvme list' → /dev/nvme0n1 (1TB). Latency: 0.15ms vs 0.35ms iSCSI on same infrastructure — 57% improvement from protocol upgrade alone.

Q29 What is FlashArray VAAI support for VMware?

A VAAI primitives on FlashArray: (1) XCOPY/Full Copy — array-side copy for Storage vMotion (zero ESXi network traffic). (2) Write Same/Zero — fast thick VMDK zeroing. (3) ATS — VMFS metadata locking without SCSI reservations. (4) UNMAP — space reclamation from deleted VMs. All VAAI operations automatically used — no admin config. Requires FC or iSCSI, host type set to ESXi in Pure host object.

Example: Storage vMotion 500GB VM without VAAI: 8 minutes, ESXi network saturated. With VAAI XCOPY: 12 seconds, zero ESXi network. VAAI ATS: 1000 VMs on same VMFS datastore — no SCSI-2 reservation lock contention. UNMAP: deleted VMs' space returned to array. VAAI is the most impactful Pure+VMware integration feature.

Q30 What is Pure vSphere Plugin?

A Pure Storage vSphere Plugin extends vCenter with Pure management. Features: create/expand/delete Pure volumes from vCenter, snapshot VMs using Pure snapshots (instant), clone VMs via Pure snapshot (thin copy in seconds), view capacity/performance from vCenter. Install: VMware Marketplace or OVA. Enables VMware admins to manage Pure storage without learning Pure CLI. Complements standard vCenter datastore management with Pure-native operations.

Example: Without plugin: VM clone = vCenter full copy (500GB transferred). With Pure plugin: VM Clone → Pure Clone → thin copy in 30 seconds, 2GB initially used (metadata only). 20 dev VMs from one 100GB template: 20 × 100GB provisioned = 2TB, physical 2.1GB (sharing parent blocks). Pure plugin makes VMware operations dramatically more efficient and faster.

Q31 What is FlashArray volume QoS?

A FlashArray QoS (Purity 5.0+): per-volume IOPS and bandwidth limits. Configure: Storage ▫ Volumes ▫ select ▫ QoS. Limit types: IOPS/sec, MB/s bandwidth. Limits only (no minimum guarantee). Prevents noisy neighbor from consuming all array IOPS. Apply to individual volumes. Monitor: Pure1 shows per-volume performance to identify throttling needs. Use case: limit batch backup, analytics, dev workloads during business hours.

Example: Analytics consuming 400K IOPS, Oracle latency spikes to 5ms. QoS: analytics_vol → limit 50K IOPS. Analytics throttled. Oracle: latency drops to 0.3ms (350K IOPS available). Production SLA maintained. QoS fixes noisy neighbor in 60 seconds — no application restart, no volume migration.

Q32 What is FlashArray space reclamation (UNMAP)?

A FlashArray receives SCSI UNMAP from hosts when files deleted. Marks physical blocks free on thin volumes. VMware: automatic UNMAP (vSphere 6.5+). Linux: 'fstrim' or 'discard' mount option. Windows: automatic TRIM (NTFS). UNMAP immediately frees physical space, improving data reduction ratio. Monitor: watch volume physical usage decrease after large delete operations. Enable on host side — FlashArray processes automatically.

Example: 50 VMs deleted from VMFS (1.8TB). Without UNMAP: FlashArray shows 3.8TB used. With UNMAP: VMware VAAI sends UNMAP for deleted VM blocks → FlashArray reclaims 1.8TB → physical drops to 2TB. Aggregate data reduction improves. Regular UNMAP = accurate capacity accounting.

Q33 What is FlashArray snapshot restore?

A Snapshot restore options: (1) Copy snapshot to new volume — non-destructive, creates new writable volume. (2) Overwrite existing volume — reverts to snapshot state (instant, old data replaced). (3) Mount snapshot read-only — connect snapshot to host for file extraction. Auto-snapshot created before overwrite for safety. CLI: 'purevolume copy --overwrite vol.snap target_vol'. Instant operation regardless of volume size.

Example: App upgrade at 10 PM corrupts database. Pre-upgrade snapshot 'oracle_data.pre-upgrade'. Rollback: GUI → snapshot → Overwrite → oracle_data. Instant revert. Oracle restarts. Total rollback: 5 minutes (mostly app restart). Pre-upgrade snapshots are standard practice on FlashArray — always take before changes.

Q34 What is FlashArray volume group management?

A Volume Groups (VGroups): logical collection of volumes managed together. All volumes in VGroup snapped simultaneously — crash-consistent across all members. Use case: Oracle (data, redo, undo, temp), SQL Server (data, log), SAP HANA (data, log). Create: Storage ▯ Volume Groups ▯ +. Add volumes. VGroup snapshot: single operation snaps all members atomically. All must be protected together for application consistency.

Example: Oracle: oracle_data, oracle_redo, oracle_undo, oracle_temp → VGroup 'oracle_prod'. Protection group: VGroup snapshot every 15 minutes. DBA drops table: restore from VGroup snapshot — all 4 volumes restored consistently. Individual volume snapshot would risk split-transaction corruption across volumes. VGroup ensures consistent multi-volume recovery.

Q35 What is FlashArray LUN masking and access control?

A Access control via host-volume connections: volume only visible to hosts explicitly connected. Host object has server's WWPNs/IQNs/NQNs. Volume connection grants access. Host Group: multiple hosts share access. No separate LUN masking layer (unlike traditional storage groups + LUN masking). Zone (FC switch) + volume connection (Pure GUI) = complete access control. Unauthorized host: no volume connection = cannot see volumes even if network/FC access exists.

Example: Traditional: storage group + zone + igroup = 3 configurations per host. Pure: zone + volume connection = 2 steps. New host: create host object, add to host group, inherits all group volume connections. 2 steps vs 5+ steps traditional. Security: without Pure host object connection, volumes invisible even if FC-zoned.

Q36 What is FlashArray data-at-rest encryption?

A FlashArray encrypts all data at rest using AES-256 — always on, zero configuration. Key management: (1) Internal — FlashArray manages keys (default). (2) External KMIP — keys on HashiCorp Vault, Thales, or IBM SKLM. FIPS 140-2 validated. Drive removal: DFMs unreadable without array's key — safe disposal. Zero performance impact (hardware-accelerated). No admin action needed to enable encryption.

Example: Healthcare PCI/HIPAA compliance: 'Are arrays FIPS compliant?' Yes — FIPS 140-2 Level 1. External KMIP: 'security key-manager external add --server vault.hospital.com:5696'. Keys on Vault, data on FlashArray — separated. DFM removed: unreadable without Vault key → safe disposal, zero data breach risk. Encryption compliance achieved with zero performance impact.

Q37 What is FlashArray stretched volume and Pod?

A Pod: FlashArray construct enabling ActiveCluster synchronous stretch. Pod contains volumes that replicate synchronously between two arrays. Volumes in a Pod appear on both arrays. ActiveCluster: Pod stretched across two arrays — writes to either replicated synchronously to partner. Mediator: arbitrates split-brain. CLI: 'purepod create pod_name', then stretch to second array. All volumes in Pod fail over together as atomic unit.

Example: 'production_pod' with 10 volumes (Oracle, SQL, VMware datastores). FlashArray A DC1 + FlashArray B DC2. DC1 write → synchronous to DC2 before ACK. DC1 failure: DC2 promotes pod volumes automatically (Mediator confirmed DC1 unreachable). DC2 hosts serve all I/O. Zero data loss. After DC1 recovery: pod resyncs, active-active restored.

Q38 What is FlashArray diagnostic and health monitoring?

A Health monitoring: GUI ▯ Dashboard (health, capacity, performance, alerts). Array ▯ Hardware (component health: controllers, PSUs, fans, DFMs). Messages ▯ Alerts (active alerts). Pure1: phone-home telemetry, ML anomaly detection. Support bundles: GUI ▯ Support ▯ Diagnostics. Monitoring integrations: vCenter plugin, Grafana (Prometheus exporter), Splunk, SNMP, PagerDuty webhooks. Pure Support: proactive case creation from Pure1 before customer notices.

Example: 3 AM: 'DFM CT0.DFM2 degraded'. Pure1 auto-notifies support. Engineer reviews remotely. Support calls at 8 AM: 'We detected an issue overnight. Field engineer scheduled for 2 PM today.' DFM replaced at 2 PM. Zero customer data loss or performance impact. Proactive support = highest customer satisfaction in enterprise storage.

SECTION 03 | REPLICATION, DR & DATA PROTECTION

Q39 What is FlashArray asynchronous replication?

A Async replication: periodic snapshot copies from source to target FlashArray. Configuration: replication connection between arrays, add target to Protection Group, set schedule (5 minutes to daily). Source creates snapshot □ transfers changed blocks to target (incremental). Target: read-only until promoted. RPO = replication interval. Monitor: Pure1 replication lag. After initial baseline, only changes transferred.

Example: Oracle NYC → Chicago async, 5-minute interval. Pure1: replication lag 3 minutes (within 5-min RPO). NYC fails: Chicago promotes PG ('purepgroup promote'). Volumes read-write. Oracle starts. RTO: 15 minutes. RPO: max 5 minutes data loss.

Q40 What is FlashArray ActiveCluster synchronous replication?

A ActiveCluster: every Pod volume write synchronously written to both arrays before acknowledgment. Zero RPO. Both arrays serve I/O (active-active). Write: local array + partner array + ACK. On site failure: surviving array continues 100% I/O, zero data loss. Mediator: third-site tiebreaker for split-brain. Requirements: <10ms RTT, Mediator, Purity 5.0+. Hosts connected to both arrays via multipath.

Example: Trading platform NYC+DC2, ActiveCluster. Every transaction on both sites. NYC earthquake destroys DC: DC2 has every transaction. Trading resumes from DC2, zero data loss. Mediator confirms NYC unreachable → DC2 promotes pod → trading continues. RPO=0 — no regulatory data loss obligation triggered.

Q41 What is FlashArray ActiveDR configuration?

A ActiveDR (Purity 6.1+): async with always-ready DR volumes. Config: replication connection + ActiveDR-enabled Protection Group + volumes in PG. DR array: demoted shadow volumes (read-only, auto-refreshed per RPO). Hosts at DR pre-configured (connected to demoted volumes). Failover: 'purepgroup promote' □ 30 seconds □ volumes writable. RTO <5 minutes. DR hosts start immediately.

Example: SQL Server NYC→Chicago, ActiveDR, 5-min RPO. Chicago: SQL volumes as demoted shadows (refreshed every 5 min). Failover: promote in 30 seconds. Chicago SQL hosts: volumes writable. SQL starts: 3 minutes. Total RTO: 4 minutes. Traditional async failover: 15+ minutes. Pre-staged hosts = key RTO advantage.

Q42 What is FlashArray replication bandwidth throttling?

A Replication throttle prevents WAN saturation. Configure: GUI □ Storage □ Arrays □ target array □ Replication Throttle □ set max MB/s. Schedule: different limits per time window (lower business hours, full speed nights). Monitor via Pure1 replication bandwidth. Initial baseline: most bandwidth needed — throttle accordingly. Delta-based after baseline: only changed blocks.

Example: 1Gbps WAN shared with production traffic. Unthrottled replication: 80Mbps consumed → network congested. Throttle: 50MB/s business hours (leaves 550Mbps for production), unlimited nights. Daily change 200GB: off-peak 400Mbps → transfers in 67 minutes. Throttling balances replication and production network coexistence.

Q43 What is FlashArray SafeMode configuration?

A SafeMode: requires Pure Support to enable. Once on: Snapshot retention periods immutable — cannot be reduced or deleted before expiry. Override requires 3-factor auth (array admin + Pure Support + Pure CSM). SafeMode per-protection-group retention periods. GUI: SafeMode indicator on protected snapshots. Eradication: 24-hour delay even after retention expiry. Designed specifically to defeat ransomware that compromises admin credentials.

Example: Enable SafeMode: call Pure Support → joint auth session → SafeMode on 'prod_pg', 72-hour retention. Ransomware attack: encrypts volumes, tries 'purepgroup destroy --eradicate prod_pg.snap'. Fails: 'SafeMode protection active.' Clean 72-hour-retained snapshot available for recovery. Three-factor requirement prevents insider and external attacker from disabling protection.

Q44 What is FlashArray CloudSnap?

A CloudSnap: replicates FlashArray snapshots to cloud object storage (AWS S3, Azure Blob, GCS). Cost-effective long-term retention. Config: add cloud credentials, create S3 bucket, add cloud target to Protection Group, set CloudSnap schedule. First run: encrypted baseline uploaded. Subsequent: incremental (changed blocks). Restore: download from cloud to FlashArray, create volume. Encrypted in transit and at rest.

Example: Primary FlashArray → Secondary FlashArray (SafeMode) → AWS S3 (CloudSnap, 90-day retention). Ransomware destroys both on-prem arrays. AWS S3: immutable objects. New FlashArray provisioned → CloudSnap restore from S3. 90-day recovery history. 90 daily snaps × ~50GB delta = 4.5TB × $0.023/GB/month = $103/month vs $50K+ secondary array. CloudSnap: 10× cheaper for archival.

Q45 What is FlashArray snapshot restore workflow?

A Restore options: (1) Copy snapshot to new volume (non-destructive, new writable volume from snapshot). (2) Overwrite existing volume (reverts to snapshot state instantly). (3) Mount snapshot read-only (connect directly for file extraction). Auto-snapshot created before overwrite for safety. GUI: Snapshots □ select □ Copy or Overwrite. CLI: 'purevolume copy --overwrite vol.snap target'.

Example: Dev deletes app config from production at 3 PM. PG snapshot from 2 PM has correct config. Copy snapshot → 'prod_recovery' volume. Connect to admin workstation. Copy specific file back to production. Delete recovery volume. Total: 8 minutes. Targeted file recovery — no full volume restore needed.

Q46 What is FlashArray replication failover testing?

A DR test without disrupting live replication: promote a historical snapshot (not current), not the live PG. Creates read-write volumes from point-in-time. Connect to DR test hosts. Test application. Delete test volumes. Active replication continues unchanged. CLI: 'purevol copy prod_pg.daily.7daysago.oracle_data oracle_dr_test'. Tests DR process without impacting live RPO.

Example: Quarterly DR test: Chicago array. Select 7-day-old snapshot. Copy to test volumes. SQL Server starts with 7-day-old data — validates DR process. Delete test volumes. Live 5-minute RPO replication untouched throughout. Zero production impact. Business owner signs off: 'DR process validated, RTO 18 minutes achieved.'

Q47 What is FlashArray pod failover and failback?

A ActiveCluster pod failover: automatic if Mediator configured (site A fails □ Mediator □ site B promotes). Manual: 'purepod failover pod_name'. Failover: volumes were already active-active, no promotion delay — site B continues immediately. Failback: site A recovered □ pod resyncs from site B (changed blocks only) □ both active-active again □ site A hosts reconnect to local Optimized paths.

Example: NYC power failure (ActiveCluster pod). Mediator: NYC unreachable → Chicago promotes automatically in 15 seconds. NYC hosts reconnect to Chicago via backup link. Chicago serves 100% I/O. NYC restored after 4 hours: FlashArray rejoins pod, 4-hour delta resyncs (45 min). Both sites active-active. NYC hosts rebalance to local Optimized paths.

Q48 What is FlashArray protection group retention design?

A Retention policy design: balance granularity, regulatory requirements, space efficiency. Example policy: every-15min/8hrs (intraday), hourly/48hrs (recent), daily/30days (monthly recovery), weekly/26weeks (semi-annual compliance). Local: recent snapshots. Remote: longer retention. CloudSnap: years for compliance. Design: start with business recovery SLA, work backwards to frequency and retention.

Example: SAP HANA: every-15min/8hrs (HANA consistency), daily/30days (operational), weekly/26weeks (compliance). Local FlashArray: 15min + daily. Remote FlashArray: daily + weekly (365-day). CloudSnap: monthly to S3 (7 years, regulatory). Each tier covers different recovery scenario at appropriate cost and RTO.

Q49 What is FlashArray replication connection setup?

A Setup: GUI □ Storage □ Arrays □ + Connect Array. Enter target array management IP + API token. Connection established. Add target to Protection Group: Protection Groups □ select PG □ Targets □ + Add Array. Enable schedule. Prerequisites: TCP port 8765 reachable. Initial baseline: all data transferred. After baseline: incremental only (changed blocks). Multiple connections possible (fan-out, fan-in topologies).

Example: NYC to Chicago: GUI → Connect Array → enter chicago_array IP + API token → connected (green). NYC PG 'prod_pg' → Targets → Add Chicago. Schedule: every 5 minutes. Baseline: 24TB over 8 hours. After baseline: daily increment 200GB in 45 minutes. Pure1 shows Chicago lag: 3 minutes average.

Q50 What is FlashArray multi-site replication topology?

A Supported topologies: (1) One-to-one: source □ single DR target. (2) One-to-many: source □ multiple targets (fan-out for multiple DR sites). (3) Many-to-one: multiple sources □ single target (backup consolidation). (4) Cascade: A □ B □ C. (5) ActiveCluster: bidirectional sync A □ B. All via Protection Groups and Replication Connections. Pure1: shows replication topology map.

Example: Finance multi-site: NYC Primary → NJ (ActiveCluster, RPO=0) + NYC → Chicago (async, 24hr RPO, 365-day retention). NYC fails: NJ takes over (<30 sec, ActiveCluster). NJ fails: Chicago archive has year of snapshots (long RTO, compliance recovery). Three-tier protection from one source using fan-out.

Q51 What is Pure Storage Mediator for ActiveCluster?

A Mediator: lightweight service on third site (independent of both array sites). Arbitrates split-brain: when Site A and B lose connectivity, both contact Mediator. Mediator decides which site continues. Prevents both sites from simultaneously serving diverged I/O (split-brain). Options: Linux VM on third site, or Pure1 cloud Mediator (simplest — Pure manages). Network: both sites must reach Mediator independently via separate paths.

Example: NYC + Chicago + AWS VPC Mediator. NYC-Chicago link fails. Both arrays contact Mediator. Mediator: 'Both can reach me — pre-configured: NYC wins.' Chicago pauses I/O. NYC continues. When link restored: Chicago resyncs from NYC. Mediator prevents data divergence in split-brain. Cloud Mediator (Pure1) eliminates need for customer-managed third site.

SECTION 04 | VMWARE, KUBERNETES & APPLICATION INTEGRATION

Q52 What is Pure Storage for VMware — key integrations?

A Pure Storage VMware integrations: (1) vSphere Plugin — manage Pure from vCenter (provision, snapshot, clone). (2) VAAI — hardware acceleration (XCOPY, ATS, Write Same, UNMAP). (3) vVols — per-VM storage policies and granular management. (4) VASA Provider — storage policy-based management. (5) Site Recovery Manager (SRM) — automated DR orchestration. (6) vRealize Orchestrator — automation workflows. Most impactful: VAAI (performance) + vSphere Plugin (simplicity) + vVols (granularity).

Example: VMware admin workflow with Pure integrations: create datastore (vCenter plugin), storage vMotion (VAAI XCOPY — 30 seconds vs 8 minutes), snapshot before upgrade (vCenter right-click → Pure Snapshot), clone dev VMs (vCenter plugin → Pure clone — 20 VMs in 60 seconds). All within vCenter — no Pure GUI or CLI knowledge required for daily VMware operations.

Q53 What is Pure Storage vVols for VMware?

A vVols (Virtual Volumes): each VM gets dedicated Pure volume instead of shared VMFS. Benefits: per-VM snapshot policy, per-VM QoS, per-VM replication, per-VM storage class. VASA Provider (Pure plugin in vCenter): mediates between vSphere and Pure. Storage Policy Based Management (SPBM): assign gold/silver/bronze policies to VMs. Protocol Endpoint (PE): special Pure volume that mediates vVol I/O.

Example: 500 VMs with vVols: DBA Oracle VM → gold policy (//X, 5-min snapshot, replication). Dev VM → bronze policy (//C, weekly snapshot). Policy assignment automatic — VM inherits protection on creation. Individual VM recovery: vCenter → restore from Pure snapshot (VM only, not entire datastore). Per-VM granularity impossible with shared VMFS.

Q54 What is Pure Storage for Kubernetes (PSO/Trident)?

A Pure Service Orchestrator (PSO) / NetApp Trident: CSI driver for Kubernetes. Dynamically provisions Pure volumes as PVs. StorageClass □ Pure backend. PVC □ volume created, zoned, host object created, connected. Pod mounts storage. PVC deletion □ volume deprovisioned. RWO: FC/iSCSI LUN. RWX: FlashBlade NFS. VolumeSnapshot API □ Pure snapshots. Helm chart deployment. StatefulSets get persistent storage automatically.

Example: StatefulSet MySQL with PVC (storageClass: pure-block). PSO creates 500GB Pure volume, zones FC, host object with node IQN, connection. MySQL mounts /data. Pod rescheduled: PSO updates host connection to new node, mounts same volume. 'kubectl apply volumesnapshot.yaml' → Pure snapshot. New PVC from snapshot → Pure clone. Full Kubernetes-native storage lifecycle.

Q55 What is Pure Storage for Oracle Database?

A Oracle on Pure best practices: (1) Separate volumes: data, redo, archive, temp. (2) VGroup for consistent snapshots. (3) ASM on raw Pure volumes. (4) RMAN + Pure snapshot integration. (5) 8KB Oracle block aligns well with Pure compression. (6) Pre-upgrade snapshot (instant rollback). (7) ActiveDR for near-zero RTO. (8) No QoS unless noisy neighbor. Pure snapshots: instant consistent backup vs hours RMAN.

Example: Oracle 19c RAC on Pure //X: VGroup 'oracle_rac' — data (2TB), redo (500GB), undo (500GB), voting (10GB). ASM diskgroups. PG: VGroup snap every 15 minutes. Pre-patching: 'oracle_rac.pre-patch'. Patch fails: overwrite all 4 volumes from pre-patch (30 seconds). Oracle RAC restored. Rollback that took 4 hours traditionally: 5 minutes on Pure.

Q56 What is Pure Storage for SQL Server?

A SQL Server on Pure: (1) FC or iSCSI connectivity. (2) Separate volumes: data, log, TempDB. (3) NTFS 64KB allocation unit for data volumes. (4) VSS integration: Pure FlashArray VSS Provider — application-consistent SQL snapshots. (5) WSFC shared volumes supported (SCSI PR). (6) AlwaysOn AG: separate Pure volumes per replica. SnapCenter-equivalent: Pure VSS snapshot + replication. SQL backup window: seconds.

Example: SQL 2019 on Pure: sql_data (1TB), sql_log (200GB), sql_tempdb (100GB). Pure VSS Provider: VSS-consistent daily snapshot (both data and log simultaneously). Replicated to DR. Retention: 30 days. Recovery: mount snapshot as new volume → attach SQL files → extract specific table → 5-minute granular recovery. SQL backup window: 30 seconds vs 3 hours traditional backup.

Q57 What is Pure Storage for VDI (Virtual Desktop)?

A VDI on Pure: VMware Horizon/Citrix on FlashArray. Key: (1) 10-20:1 deduplication (Windows OS blocks shared across all VMs). (2) Instant VM provisioning via Pure snapshot clone. (3) Boot storm: DRAM cache absorbs burst — no performance degradation. (4) VAAI ATS: VMFS locking without contention for thousands of VMs. (5) QoS: per-VM group limits. Best platform for high-density VDI due to extreme dedup ratios.

Example: 1000 Windows 10 VDI sessions on //X50: data reduction 8.5:1. Boot storm 1000 VMs simultaneously: FlashArray DRAM cache serves repeated OS blocks — 600K IOPS at <0.5ms. Steady state: 150K IOPS, 0.2ms. Physical: 1000 × 50GB = 50TB provisioned → 5.9TB physical. Pure VDI = best dedup ratio + boot storm resilience.

Q58 What is Pure Storage for SAP HANA?

A SAP HANA on Pure: SAP TDI certified, SAP Note 2399728. Requirements: //X (performance), NVMe/FC or FC, dedicated data + log volumes. HANA Backint plugin: Pure provides plugin □ HANA backup API triggers Pure snapshot (instant, consistent). SnapCenter for SAP orchestrates Backint + Pure. HANA HSR (System Replication) + Pure async = complete DR. Certified for HANA TDI = full SAP support.

Example: S/4HANA 4TB on Pure //X70: data volume 4.8TB, log 600GB. Backint: HANA triggers Pure snapshot → consistent backup in 5 seconds. Recovery: Backint restores from snapshot in 5 minutes vs 3-hour traditional. ActiveDR: 5-minute RPO for HANA DR. SAP TDI certified — Pure fully supported by SAP for HANA workloads.

Q59 What is Pure Storage FlashArray SRM integration?

A VMware Site Recovery Manager (SRM) with Pure: Pure Storage SRA (Storage Replication Adapter) integrates with SRM for automated DR orchestration. SRM discovers Pure replication relationships. DR test: SRM mounts replicated volumes as test snapshots (non-disruptive). Planned migration: SRM coordinates VM shutdown + Pure failover + VM restart. Unplanned failover: SRM promotes Pure replication + restarts VMs. GUI-driven DR runbook execution.

Example: SRM DR test: click 'Test Recovery Plan' in vSphere. SRM: pause Pure replication briefly, mount DR snapshot on DR hosts. SRM starts VMs from snapshot (non-disruptive to production replication). Validate VMs, network, application. SRM cleanup: unmount test snapshots, resume replication. Zero production impact. DR test in vSphere GUI — storage and compute orchestration unified.

Q60 What is FlashArray Purity upgrade procedure?

A Upgrade: (1) Pure1 evaluates readiness. (2) Download package. (3) Pre-check: 'pureupgrade pre-upgrade-check'. (4) 'pureupgrade upgrade'. (5) CT1 evacuates to CT0, CT1 reboots new Purity (~8-10 min). (6) CT0 evacuates to CT1, CT0 reboots (~8-10 min). Complete: 20 minutes, 2 path flaps. MPIO required on all hosts. Schedule during low traffic. Post-upgrade: verify paths, volumes, replication health.

Example: Saturday 2 AM upgrade: CT1 failover at 2:00 (I/O on CT0, 10 min). CT1 returns at 2:10. CT0 failover at 2:12 (I/O on CT1, 10 min). CT0 returns at 2:22. Both upgraded. VMware: DRS/HA functional throughout. SQL: MPIO maintained, zero errors. Recommended: verify MPIO before upgrade, document single-path period in change ticket.

Q61 What is FlashArray monitoring integrations?

A Monitoring: (1) Pure1 — primary, AI-driven, proactive. (2) Grafana: Pure Storage plugin, REST API metrics. (3) Prometheus: Pure Exporter □ Prometheus □ Grafana dashboards. (4) Splunk: Pure App — logs + metrics. (5) vCenter: Pure Plugin — performance in vSphere. (6) SNMP v3: encrypted traps. (7) Webhooks: PagerDuty, Slack, Teams. (8) ServiceNow: REST API for CMDB. Multiple overlapping monitoring channels for comprehensive coverage.

Example: Monitoring stack: Pure1 (proactive, ML) + Grafana real-time dashboards + PagerDuty webhooks (critical alerts) + ServiceNow CMDB (auto-populated). Alert: Pure1 detects DFM anomaly → webhook → PagerDuty page → ServiceNow incident auto-created. End-to-end: detection to ticket in <2 minutes, all automated.

Q62 What is FlashArray REST API for automation?

A REST API 2.x automation patterns: volume lifecycle (create/connect/delete), snapshot management, host management, protection group operations, performance data collection, capacity reporting. Python: py-pure-client. PowerShell: purestorage module. Ansible: purestorage.flasharray collection. Terraform: purestorage provider. API enables: self-service portals, CI/CD pipeline storage provisioning, automated DR testing, capacity reporting to management.

Example: Full automated provisioning via Ansible: (1) purefa_host: create host with IQN. (2) purefa_vol: create 1TB volume. (3) purefa_hg: add host to host group. (4) purefa_connect: connect volume. ServiceNow request → Ansible → Pure API → volume provisioned in 30 seconds → ticket auto-resolved. Zero storage admin manual steps.

Q63 What is Pure Storage FlashStack (validated reference architecture)?

A FlashStack: validated reference architecture = Pure FlashArray + Cisco UCS compute + Cisco Nexus networking. Pre-tested for common workloads (VMware, Oracle, SAP, VDI). Components: Cisco UCS blades, Cisco Nexus 9000, Cisco MDS FC switches, Pure FlashArray//X or //C. Benefits: faster deployment (pre-validated), reduced design risk, single-vendor support for full stack, Cisco Intersight + Pure1 joint management.

Example: New DC: FlashStack for VMware — Cisco UCS + Pure //X50 + Nexus 9000 + MDS 9132T. Pre-validated design: FC zoning template, UCS profiles, VMware config guide — all documented. Deployment: 5 days vs 3 weeks custom. Cisco + Pure joint support. Reference architecture reduces deployment risk and time significantly.

Q64 What is Pure Storage AI and ML workload support?

A AI/ML on Pure: FlashArray//X for structured data (indexes, metadata) via NVMe/FC for lowest latency. FlashBlade for unstructured datasets (NFS/S3) — high throughput. NVMe/FC or NVMe/RoCE for GPU server connectivity. CloudSnap for dataset backup/archive. Key metric: throughput (GB/s) for sequential dataset reads. GPU utilization metric: >90% means storage not bottlenecked — target performance.

Example: NVIDIA DGX cluster: FlashArray//X90 via NVMe/FC for structured data, FlashBlade//S for NFS dataset. DGX reads 1.2TB ImageNet → //X90 delivers 300GB/s (NVMe/FC). GPU utilization: 96% (storage not bottlenecking). Traditional HDD storage: GPU utilization 40% (waiting for I/O). NVMe/FC storage: GPU efficiency increases 2.4x.

SECTION 05 | PURE1, PERFORMANCE & TROUBLESHOOTING

Q65 What is Pure1 AI analytics capabilities?

A Pure1 AI/ML analytics: (1) Global Fleet — all arrays worldwide in one browser. (2) Workload Planner — model 'what if' scenarios (add workload X, impact on performance?). (3) Capacity Optimizer — identify under-utilized arrays, consolidation opportunities. (4) Anomaly Detection — ML baseline, auto-detect deviations from normal patterns. (5) Upgrade Advisor — recommend Purity version and timing. (6) Sustainability — power consumption tracking. Powered by phone-home telemetry from all Pure fleets globally.

Example: Pure1 Workload Planner: current array 400K IOPS, 0.3ms. Add SAP HANA (200K IOPS estimated): model shows 600K IOPS at 0.45ms — within SLA. Decision in 10 minutes using ML model vs weeks of capacity analysis. Workload Planner replaces manual spreadsheet capacity planning with data-driven simulation.

Q66 What is FlashArray performance analysis?

A Performance analysis: GUI □ Analysis tab: array IOPS, throughput, latency over time. Per-volume breakdown: identify highest-consuming volumes. Queue depth: high = potential overload. Read/write ratio. Latency heatmap: identify spikes. Pure1: historical trending, anomaly detection, workload correlation. Key: correlate latency events with volume-level IOPS to identify root cause (noisy neighbor, array saturation, specific volume).

Example: Oracle reports 5ms latency (normally 0.3ms). Pure1: spikes correlate with 'analytics_job_vol' generating 800K IOPS burst. Array total near limit. QoS analytics_vol → limit 100K IOPS. Oracle latency: 0.3ms in 2 minutes. Root cause identified in 5 minutes using Pure1 correlation vs hours without correlated analytics.

Q67 What is FlashArray key performance metrics?

A Monitor: (1) Array IOPS — total R+W/sec. (2) Throughput — MB/s. (3) Latency — microseconds (R and W separately). (4) Queue depth — host queue. (5) Data reduction ratio. (6) Capacity — physical vs raw, forecast. (7) Controller CPU %. (8) Replication lag — RPO compliance. (9) Protection group snapshot compliance. Via: Pure1, Grafana/Prometheus, API polling. Set alerts: latency >2× baseline, capacity >75%, replication lag >RPO.

Example: Monthly review dashboard: IOPS trend 350K→480K (37% growth in 3 months). At rate: 1M IOPS limit in 4 months. Latency: stable 0.3ms — array not stressed yet. Capacity: 45TB/80TB — 2-year forecast. Replication lag: <3 min (SLA 5 min) — compliant. Action: plan IOPS expansion (//X90 upgrade) before latency impact.

Q68 What is FlashArray data reduction analysis?

A Data reduction: Total = Dedup ratio × Compression ratio. GUI □ Dashboard □ Data Reduction section. Per-volume: Storage □ Volumes □ columns for Unique, Shared, Data Reduction. Factors: uncompressible (encrypted) data reduces savings, high-change workloads reduce dedup, similar data (VDI, VMware) maximizes dedup. Pattern elimination: zeros not stored at all. Typical: VMware 5-10:1, Oracle 2-3:1, VDI 10-20:1.

Example: Pure dashboard: Total 5.7:1. Dedup 3.1:1, Compression 1.8:1. 200TB provisioned, 64TB unique, 136TB shared, 35TB physical. Finance: 'Do we need more storage?' Answer: 165TB provisioned capacity remaining (thin). Physical expansion needed when approaches 75TB (18 months at current rate). Data reduction drives procurement decisions.

Q69 What is FlashArray NVMe performance advantages?

A NVMe architecture advantages: (1) Eliminated SSD controller: -50-100μs latency. (2) 65535 NVMe queues vs SCSI single queue — massive parallelism. (3) Global NAND management: no per-SSD garbage collection. (4) NVMe/FC connectivity: additional -50-100μs vs SCSI FC. (5) Consistent sub-0.5ms latency. (6) Higher IOPS/watt efficiency. End-to-end NVMe: NVMe host protocol + DirectFlash NVMe modules = minimum latency path.

Example: Same workload //X90 NVMe/FC vs legacy FC-SCSI: NVMe/FC: 8M IOPS, 0.15ms. FC-SCSI: 2M IOPS, 0.4ms. 4× IOPS, 63% lower latency. AI training 6hrs → 2.5hrs. Real-time analytics query 12sec → 3sec. NVMe end-to-end delivers fundamental improvement vs legacy SCSI architecture.

Q70 What is FlashArray troubleshooting methodology?

A Troubleshoot: (1) Pure1 — anomaly alerts, baseline comparison. (2) Array dashboard — current IOPS, latency, capacity. (3) Per-volume analysis — highest consumers. (4) Pure messages log — hardware and system events. (5) Host side — MPIO paths, host I/O patterns. (6) Phone-home bundle: 'purearray phonehome create'. (7) REST API: GET /api/2.x/arrays/performance for real-time. (8) Pure Support: proactive, often diagnoses before customer calls.

Example: Latency 0.3ms → 1.2ms. Troubleshoot: Pure1 no hardware alerts. Dashboard: IOPS 900K (near 1.1M max). Per-volume: backup_job 450K IOPS. Root cause: backup during business hours. Fix: QoS backup_vol → 50K IOPS. Latency: 0.3ms in 2 minutes. Total time: 8 minutes with Pure1 + GUI analytics.

Q71 What is FlashArray read/write cache?

A Cache: Write cache = NVRAM in each controller. Writes acknowledged after both controllers NVRAM confirm — safe from controller failure. NVRAM flushes to DFMs continuously. Read cache = DirectMemory Cache (DMC) — DRAM for hot data. DMC serves hot data at DRAM speeds (<0.1ms). Cache hit ratio visible in GUI. No separate SSD cache tier — Purity manages DRAM + NVRAM + DirectFlash as unified intelligent cache.

Example: Oracle OLTP hot indexes: DMC hit rate 85% → 85% reads served from DRAM (<0.1ms). 15% cache miss → DirectFlash (0.2ms). Average read: 0.12ms. No admin tuning needed — DMC auto-caches hot data. No performance cliff when data exceeds DMC — DirectFlash consistent 0.2ms.

Q72 What is FlashArray proactive support (phone-home)?

A Phone-home: FlashArray sends telemetry to Pure1 via HTTPS continuously. Data: performance metrics, hardware health, errors, configuration. Pure1 ML: detect anomalies vs global peer fleet baselines. Predictive: DFM, PSU, fan wear prediction before failure. Auto case creation: Pure Support opens case before customer notices. Remote diagnosis: Support accesses array state via Pure1 (with consent). Requires: HTTPS outbound to Pure1 cloud.

Example: Pure1 ML detects unusual write amplification on CT0.DFM5 — statistically anomalous vs 10K similar DFMs. Auto-alert to Support. Engineer calls customer: 'We detected DFM anomaly, predict failure in 3 weeks. Field engineer Tuesday?' DFM replaced, zero failure occurs. Customer: 'We didn't notice anything.' — This proactive model is Pure's #1 differentiator from competitors.

Q73 What is FlashArray capacity expansion procedure?

A Expansion: (1) Monitor: approaching 75% physical □ order DFMs 8 weeks advance. (2) Pure Support or pure1.purestorage.com: submit order. (3) Field engineer: delivers + installs DFMs. (4) Hot-plug: slide DFMs in available bays. (5) Purity: auto-detects, integrates capacity immediately. (6) Rebalancing: Purity redistributes data across DFMs for even wear (background, non-disruptive). (7) New capacity available instantly for provisioning.

Example: //X50 at 70% (35/50TB). Order 12 DFMs (+36TB). Field engineer: inserts 12 DFMs in 30 minutes (hot-plug). Array: 86TB raw immediately. Background rebalancing: 6 hours, no app impact. After rebalance: all DFMs equal distribution. Zero downtime. Evergreen//Forever: DFM expansion included in subscription — no separate purchase needed.

Q74 What is FlashArray benchmark methodology?

A Benchmark: (1) FIO (Linux) or DiskSPD (Windows). (2) Warm up: sequential writes to prime flash. (3) Test: randread 4KB (IOPS), randwrite 4KB, sequential 1MB (throughput). (4) Queue depths: 1, 8, 32, 128 — find optimal. (5) Realistic workload: not 100% zeros (artificially compressible). (6) Monitor Pure GUI simultaneously — verify array-side metrics. (7) Document: IOPS, throughput, P50/P99/P999 latency.

Example: FIO on //X50: 'fio --ioengine=libaio --direct=1 --rw=randread --bs=4k --numjobs=8 --iodepth=32 --size=100g --runtime=60'. Result: 1.48M IOPS, 0.17ms. Pure spec: 1.5M IOPS — 99% of spec achieved. Write: 800K IOPS, 0.3ms. Mixed 70/30: 1.1M IOPS, 0.22ms. All match Pure SPECsfs benchmarks. Baseline documented for future regression testing.

Q75 What is FlashArray upgrade planning?

A Annual upgrade planning: (1) Capacity: 15%/year growth □ expansion needed in 2 years. Order DFMs 8 weeks lead time. (2) Controller: Evergreen//Forever □ new gen every 3 years (non-disruptive). (3) Purity software: Pure1 Upgrade Advisor recommends version and timing. Check IMT for host OS/HBA compatibility. Test environment first. (4) Schedule: low-traffic window for each controller reboot (~10 minutes). (5) Post-upgrade: verify paths, volumes, replication.

Example: Annual planning: Physical 65%, 15%/year growth → order DFMs in 4 weeks. Purity: Pure1 recommends 6.1.4 from 5.3.12. IMT: ESXi 8.0 + Purity 6.1.4 — supported. Schedule Saturday 2 AM. Pre-check passes. Upgrade: 20 minutes. Post: all paths verified, volumes healthy, replication lag normal. Document in change management.

SECTION 06 | ADVANCED ADMINISTRATION & BEST PRACTICES

Q76 What is FlashArray CLI overview?

A CLI via SSH to management IP. Commands: purevol (volumes), purehost (hosts), purehgroup (host groups), puresnap (snapshots), purepgroup (protection groups), purearray (array info), pureport (ports), purenetwork (network), puremessage (alerts). Help: 'purevol --help'. Output: table or --csv. Login: 'ssh pureuser@array_ip'. All GUI operations available via CLI. REST API preferred for automation; CLI for ad-hoc and troubleshooting.

Example: 'purevol list' (all volumes). 'purevol create --size 1T oracle_data'. 'purehost create --iqnlist iqn.xxx:host01 host01'. 'purehgroup create --hostlist host01,host02 vmware_cluster'. 'purevol connect --hgroup vmware_cluster oracle_data'. 'puresnap list oracle_data'. 'purearray monitor' (performance). 'puremessage list --flagged' (active alerts).

Q77 What is FlashArray RBAC?

A Built-in roles: (1) Array Admin — full access. (2) Storage Admin — volumes, hosts, protection groups (no array-level config). (3) Operations Admin — operations, no destroy/eradicate. (4) Read Only — view only. LDAP/AD: map AD groups to Pure roles. Audit: 'puremessage list --audit' — all admin actions with user + timestamp. No custom roles. Principle of least privilege: storage team □ storage_admin, backup team □ ops_admin, helpdesk □ read_only.

Example: Role assignment: Storage Admins → storage_admin (provision, delete volumes). Backup team → ops_admin (manage PGs, snapshots, no volume delete). Helpdesk → read_only (view performance, capacity). Array changes (network, replication, upgrade) → array_admin (2-person team). Monthly audit: 'puremessage list --audit' — all actions logged with AD username.

Q78 What is FlashArray network troubleshooting?

A Troubleshoot: (1) Ping: 'purearray ping --target 10.200.0.1'. (2) iSCSI ports: 'pureport list --iscsi' shows IPs + state. (3) FC ports: 'pureport list --fc' shows WWPNs + state. (4) Network config: 'purenetwork interface list'. (5) Host side: ping storage IP, check VLAN tagging, firewall rules. (6) FC: 'show flogi database vsan 10' — array WWPNs logged in? (7) iSCSI sessions: 'iscsiadm -m session' — sessions to array?

Example: ESXi cannot see Pure volumes after network change. Pure GUI ports: all green. Host: 'esxcli iscsi session list' — no sessions to Pure IPs. Ping: ESXi vmkernel cannot ping Pure iSCSI IP. Network: VLAN 200 not tagged on ESXi vmkernel after switch upgrade. Fix: add VLAN 200 tagging. Ping succeeds. Sessions establish. Volumes visible. VLAN misconfiguration — not storage issue.

Q79 What is FlashArray FC troubleshooting?

A FC troubleshoot: (1) 'pureport list --fc' — array FC WWPNs and state (online?). (2) Switch: 'show flogi database vsan 10' — array WWPNs logged in? (3) Zone: array WWPN in zone with host WWPN? (4) Host: HBA online? (systool -c fc_host -v on Linux). (5) 'multipath -ll' — paths to Pure volumes? (6) Pure host object: correct host WWPNs? (7) Volume connection: volume connected to host in Pure GUI?

Example: New Linux cannot see Pure via FC. Pureport: FC online. Flogi DB: array present. Zone check: host WWPN NOT in zone with array WWPN! Create zone: host WWPN + Pure CT0.FC0 WWPN. Activate zoneset. Multipath shows 2 paths. /dev/sdb appears. Root cause: missing FC zone — #1 FC connectivity failure reason.

Q80 What is FlashArray snapshot management and eradication?

A Snapshot lifecycle: Create □ Retained □ Destroyed (soft delete, 24hr recoverable) □ Eradicated (permanent). Destroy: 'puresnap destroy vol.snap'. Recover within 24hrs: 'puresnap recover vol.snap'. Eradicate: permanent. SafeMode: blocks eradication until retention expires. Pending eradication: 'puresnap list --pending-eradication'. Protection Group auto-managed snapshots: created/deleted per retention policy.

Example: Accidental snapshot delete: 'puresnap destroy oracle_data.critical'. Recover immediately: 'puresnap recover oracle_data.critical' — restored. After 24hrs: permanently eradicated. SafeMode: destroy fails → 'SafeMode protection active' → cannot delete until retention expires. Two-stage deletion provides accidental delete protection.

Q81 What is FlashArray data migration from competitive storage?

A Migration methods: (1) Host-based: mount source + Pure simultaneously, copy (dd/robocopy/rsync). (2) VM-based: Storage vMotion for VMware. (3) App-based: backup from source, restore to Pure. (4) Pure Migration Services: Pure-provided tools. Pure often offers free migration assistance in purchase. Typical: 2-4 weeks for medium enterprise. VAAI XCOPY: if migrating between Pure datastores (same array) = near-instant.

Example: NetApp ONTAP iSCSI → Pure //X50: Create Pure volumes. Connect to same Linux hosts as ONTAP. 'dd if=/dev/sdb of=/dev/sdc bs=64M' (ONTAP → Pure, parallel for multiple LUNs). 100TB: ~2 days. VMware: Storage vMotion datastores. Post-migration: Pure data reduction 3.2:1 → same data in 68% less physical space. ONTAP decommissioned week 2.

Q82 What is FlashArray DR failover runbook?

A Failover runbook: (1) Declare disaster. (2) Verify last replication lag. (3) Promote PG: 'purepgroup promote prod_pg' (30 seconds). (4) Verify volumes writable. (5) DR hosts discover volumes (pre-configured iSCSI/FC). (6) Start applications. (7) Validate. (8) Document RTO/RPO achieved. Failback: resync replication □ promote back □ redirect hosts. Practice quarterly.

Example: T+0 disaster. T+5 verify: last snap 4 min ago (RPO 4 min). T+7 promote PG (30 sec). T+8 volumes writable. T+10 DR hosts discover (pre-staged). T+15 Oracle starts. T+20 validation passes. T+22 DR complete. RTO: 22 min (SLA 30 min). RPO: 4 min (SLA 15 min). Quarterly test validated runbook accuracy.

Q83 What is FlashArray certificate and security management?

A Security: (1) HTTPS only — replace default self-signed cert with CA cert. (2) SSH key-based auth recommended. (3) API tokens for automation (not passwords in scripts). (4) SAML 2.0 SSO (Okta, Azure AD, Ping). (5) Syslog: audit logs to SIEM. (6) SNMP v3: encrypted monitoring. (7) Management VLAN isolated from data. (8) FIPS validated. Replace self-signed: GUI □ Array □ Security □ SSL Certificates.

Example: Hardening: DigiCert SSL cert (no browser warning). SAML SSO with Azure AD (MFA enabled, no local passwords). API tokens for scripts (in HashiCorp Vault). Syslog to Splunk. SNMP v3. Management VLAN 999. Result: all admin logins via AD+MFA, audit trail in Splunk, no shared passwords, trusted SSL. Enterprise security posture achieved.

Q84 What is FlashArray capacity planning?

A Capacity planning: (1) Current physical usage + growth trend (Pure1 forecast). (2) Data reduction trend (improving? stable?). (3) Alert at 75% physical — order DFMs 8 weeks lead time. (4) Provisioned vs physical: thin provisioning enables 2-5× overcommitment safely. (5) Snapshot overhead: incremental (changed blocks only) — typically small vs primary data. (6) Replication: source and target capacity planning independent.

Example: //X50 physical 45TB/80TB (56%), growth 2TB/month. 75% threshold = 60TB → 7.5 months. Order DFMs now (8-week lead time). Add 36TB raw DFMs → 116TB total. New threshold: 87TB → 21 months before next expansion. Capacity planning prevents emergency orders (3× premium cost) and performance degradation from overfull arrays.

Q85 What is Pure Storage Evergreen//One service?

A Evergreen//One: storage-as-a-service. Pure delivers, installs, monitors, replaces hardware. Customer pays per TB consumed monthly. Scaling: need more □ notify Pure □ capacity added within SLA. Performance SLA guaranteed. OpEx model. Pure manages Purity upgrades, hardware refresh, DFM replacements — all included. Customer: focus on applications, not storage infrastructure. Ideal for: customers without storage expertise or preferring OpEx.

Example: Hospital: 'We want patient data stored reliably without managing storage.' Evergreen//One: Pure delivers FlashArray, hospital pays $X/TB/month. End of month: 220TB → invoice for 220TB. Growth: Pure adds capacity. IT: never interacts with hardware. Pure1 proactive support handles all issues. Hospital IT refocuses from storage management to patient care applications.

Q86 What is FlashArray support model?

A Pure Support tiers: Evergreen Bronze (24×7, next business day hardware). Evergreen Silver (24×7, 4-hour hardware). Evergreen Gold (24×7, 4-hour + on-site SLA). All: Pure1 proactive support (ML detects issues, auto-opens cases). Remote diagnosis via Pure1 phone-home. Chat support in Pure1 dashboard. Key: phone-home telemetry enables remote diagnosis for 90% of issues without on-site visit. Support portal: support.purestorage.com.

Example: Traditional storage support: customer calls → logs collected manually → engineer diagnoses next day. Pure: Pure1 detects latency anomaly 2 AM → auto-alert → remote diagnosis → engineer calls customer 8 AM with solution ready. Resolution: <30 minutes vs 2 days traditional. Proactive, telemetry-driven support is Pure's most differentiated service capability.

Q87 What is FlashArray reporting and compliance?

A Compliance reporting: (1) Pure1 capacity trends (historical). (2) Data reduction ROI (dedup + compression savings). (3) Replication RPO achievement over time. (4) Audit log: 'puremessage list --audit --start date'. (5) Snapshot compliance: verify PG snapshots created per schedule. (6) Pure1 export: CSV/PDF for management. (7) REST API: custom automated reports. (8) Chargeback: per-volume capacity usage via API.

Example: Monthly report via Python REST API: capacity used (56%), data reduction 5.2:1 ($280K savings vs raw cost), replication RPO compliance (98.7% within 5-min SLA), snapshot compliance (100%), top 10 volumes by IOPS. Email to management and finance for chargeback. Pure API makes comprehensive reporting straightforward without manual data collection.

Q88 What is FlashArray production deployment best practices?

A Best practices checklist: (1) Dual FC paths (CT0 + CT1 ports) or dual iSCSI. (2) MTU 9000 for iSCSI. (3) MPIO with ALUA. (4) Correct host type in Pure host object. (5) VGroups for multi-volume apps. (6) Protection Group with schedule (minimum daily). (7) Remote replication for tier-1 (minimum 5-min RPO). (8) SafeMode enabled. (9) Pure1 connected (phone-home). (10) Monitoring alerts configured.

Example: Pre-production Oracle checklist: [x] 4 FC paths (2 per CT). [x] ALUA multipath verified. [x] Host type = Linux. [x] VGroup with data/redo/undo/temp. [x] PG: 15-min snaps, 4hr retain, daily 30-day retain. [x] Remote replication every 5 min. [x] SafeMode 72-hour. [x] Pure1 phone-home verified. [x] QoS documented. All 10 checks: production ready.

Q89 What is FlashArray component replacement?

A Component maintenance: (1) DFM replacement: hot-swap, Purity handles redistribution automatically. (2) Controller: Evergreen procedure, non-disruptive, partner serves I/O during replacement. (3) Power supply: hot-swap, redundant PSUs. (4) Fan: hot-swap. (5) SFP/cable: replace while monitoring port state. (6) DFM firmware: updated during Purity upgrades. All replacements: hardware-only, no Purity reinstall. Pure1 alerts trigger before manual detection.

Example: DFM replacement: Pure alert 'CT1.DFM3 degraded'. Pure schedules replacement. Purity marks DFM3 failed, redistributes (30 min). Field engineer: removes DFM3 hot-swap. Inserts new DFM3. Purity incorporates, redistributes data back. Array: full health. Customer: zero downtime, zero data loss. DFM replacement = cleanest hardware service in enterprise storage.

Q90 What is FlashArray REST API authentication?

A API authentication: API Tokens — generate per-user via GUI □ Users □ API Token. Never expires unless revoked. Bearer token: 'Authorization: Bearer ' in requests (REST 2.x). Session token: POST /api/login □ session cookie. Rate limit: 100 req/sec per token. TLS 1.2+. Audit: API calls logged. Best practice: dedicated 'automation' user with minimum required role, token stored in secrets manager (Vault, AWS Secrets Manager).

Example: Set Authorization header with Bearer token and call REST endpoint for volumes listing. Token in HashiCorp Vault (not in script). Dedicated automation_user with storage_admin role. If compromised: revoke token — no impact on other users. API authentication: security best practices identical to any modern enterprise API.

Q91 What is FlashArray disaster recovery testing best practices?

A DR testing: (1) Test quarterly, not annually. (2) Non-disruptive: use historical snapshot, not live replication. (3) Document RTO/RPO achieved vs SLA. (4) Full application validation (not just disk visibility). (5) Automate via REST API (reduce human error). (6) Business owner sign-off. (7) Lessons learned: update runbook. (8) Practice failback — often forgotten. (9) Test multiple recovery scenarios: planned, unplanned, ransomware.

Example: Quarterly DR test: restore 7-day-old snapshot (not latest). SQL Server starts on DR volumes. Validation queries match expected results. Failback practiced. RTO: 18 min (SLA 30 min). RPO tested: 7 days (SLA 1 hour). Runbook updated: step 7 missed, added. Business owner signs off. Next test: add ransomware recovery scenario. DR insurance that actually works when needed.
